

JVM Lab: HikariCP Connection Pool Starvation

Duration : 30 Min.

Scenario

An API service under load starts throwing `ConnectionTimeout` exceptions, indicating database connection pool starvation.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Tune HikariCP parameters including `maximum-pool-size`, `checkout timeout`, and `leak-detection-threshold` configurations to stabilize pool metrics.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. HikariCP database connection pool sizing configurations: 5 marks
2. Leak detection and checkout timeout optimization check: 5 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.